

System Architecture: Generative AI Video Platform

Multi-Agent Autonomous Collaboration, LLM Reasoning Core & Closed-Loop Quality Control

1. System Overview & Cognitive Decoupling

The platform automatically transforms natural-language video ideas into broadcast-ready videos. Crucially, the **Large Language Model (LLM)** acts as the **cognitive brain and orchestrator** (performing planning, script decomposition, prompt synthesis, and quality auditing) rather than directly generating raw video pixels. Domain-specialized AI agents interface through a **Unified Model Gateway** to drive diffusion image generators, video motion models, neural TTS, and music synthesizers.

2. 6-Layer Architectural Hierarchy

Layer	Component Name	Key Responsibilities & Technologies
Layer 1	Presentation Layer	Next.js / Streamlit web interface for prompt entry, style selection (16:9/9:16), voice selection, real-time pipeline stepper, and final video player.
Layer 2	API & Application Services	FastAPI Gateway, OAuth2/JWT authentication, rate limiting, prompt sanitization, asynchronous Celery/Redis job queues, and WebSocket event streams.
Layer 3	Cognitive Orchestration (Brain)	Orchestrator Agent (DAG state coordinator) + LLM Reasoning Engine (Gemini 1.5 Pro / GPT-4o) for task decomposition, context routing, and agent synchronization.
Layer 4	Specialized AI Agent Swarm	8 Autonomous Agents: (1) Script & Story, (2) Storyboard, (3) Prompt Engineering, (4) Visual Asset, (5) Video Generation, (6) Voice & Audio, (7) Video Editor, (8) QC Critic.
Layer 5	Model Gateway & Foundation Models	Unified proxy routing to Diffusion models (Flux, SDXL), Video Gen (Gen-3, Sora, Kling), Neural TTS (ElevenLabs), and Music AI (Suno, MusicLM).
Layer 6	Data & Storage Infrastructure	Vector Database (Chroma/Pinecone) for character/style embeddings, PostgreSQL for job metadata, Redis for caching, S3/MinIO for raw clips & final MP4.

3. Specialized AI Agent Swarm & Delegation Matrix

Specialized Agent	Input Artifact	Execution Output	Target Generative Model
1. Script Agent	User prompt & tone constraints	Scene-by-scene structured dialogue & narration	LLM (Structured JSON Mode)
2. Storyboard Agent	Scene-by-scene script	Camera angles, lighting, characters & duration specs	LLM Reasoning & Rules Engine
3. Prompt Agent	Storyboard specifications	Model-optimized positive/negative prompts & seeds	LLM Prompt Optimizer
4. Visual Agent	Visual generation prompts	Consistent character reference sheets & keyframe art	Diffusion Models (Flux / SDXL)
5. Video Agent	Keyframes + video prompts	Individual high-motion video clips per scene	Runway Gen-3 / Sora / Kling
6. Audio Agent	Dialogue text + mood tags	Neural narration, SFX & adaptive music stems	ElevenLabs TTS + Suno AI
7. Editing Agent	Video clips + audio stems	Timeline cut, color grading, subtitles & final sync	FFmpeg / Automated NLE Engine
8. QC Critic Agent	Draft video + original specs	Pass/Fail score + Selective regeneration instructions	Vision-LLM (Gemini / GPT-4V)

4. End-to-End Workflow & Self-Healing QC Feedback Loop

Sequential Flow: User Prompt → API Gateway → Orchestrator Agent → LLM Planning → Script Agent → Storyboard Agent → Prompt Agent → Visual Agent → Video Agent → Audio Agent → Editing Agent → QC Critic Agent.

Closed-Loop Quality Control:

- **Acceptance Path (Score ≥ 85%):** The video meets all prompt adherence, audio-visual sync, and physical coherence thresholds → Published to Object Storage & delivered to User.
- **Rejection & Self-Healing Path (Score < 85%):** The QC Critic Agent detects flaws (e.g., motion distortion in Scene 3 or desynced voiceover) and issues a structured retry payload to the Orchestrator. The Orchestrator initiates **targeted re-execution only for the defective subtask** (avoiding expensive full-video regeneration).

5. Example Scenario: 60-Second Futuristic City Video

Step	Executing Agent / Module	Operational Workflow & Intermediate Output
1	User Input	'Create a 60-second cinematic sci-fi video of a futuristic city with flying cars and sky gardens. 16:9, deep male narration.'
2	Orchestrator + LLM	Decomposes request into 4 distinct 15s scenes: (1) Skyline Establishing Shot, (2) Maglev Highway, (3) Bio-dome Garden, (4) Horizon Outro.
3	Script Agent	Generates 4 narrative beats with precise word counts tuned for 145 wpm narration.
4	Storyboard Agent	Defines camera angles (aerial drone, tracking dolly, macro focal lens) and cyber-organic lighting parameters.
5	Prompt Agent	Generates optimized diffusion & video prompts with negative prompts for artifact reduction.
6	Visual Agent	Generates 4 master keyframe images with consistent architectural styles using Vector DB seeds.
7	Video Agent	Generates 4 high-definition video clips applying camera motion vectors (Runway Gen-3 / Sora).
8	Voice & Audio Agent	Synthesizes voiceover in ElevenLabs, generates ambient synth score, and layers futuristic vehicle sound effects.
9	Video Editing Agent	Assembles timeline, applies seamless crossfades, aligns audio stems, and renders high-contrast subtitles.
10	QC Critic Agent	Audits motion smoothness and audio alignment → Score: 93% (PASS) → Final 4K MP4 delivered to user.